Document Type: Special Notice
Posted Date: 04 May 2026
Category: Research and Development
Set Aside: N/A
**Contracting Office Address**

Department of the Army, Army Contracting Command, ACC - APG (W911NF) Research Triangle Park Division, 800 Park Office Drive, Research Triangle Park, NC 27709

**Description**

This modification references the AAL BAA which can be found at https://aal.army/assets/files/pdf/baa-published-announcement.pdf

**Synopsis:** The Army Applications Laboratory (AAL) and the Pathway for Innovation and Technology (PIT) are seeking technologies related to Expeditionary Power and Energy (power generation, energy storage, power distribution, and battery charging) to enable 11th Airborne Division (Arctic) as a Special Topic under the AAL Broad Agency Announcement (BAA) W911NF-24-S-0008 for Disruptive Applications and related research as described below. Upon receipt, the Government will evaluate submissions through the process established in the AAL BAA, except as modified herein. The Government reserves the right to cancel this special notice without award for any reason. Issuance of this notice does not commit the Government to pay for any preparation costs incurred in compiling a response.

As a general matter, all requirements referenced in BAA W911NF-24-S-0008 apply to this effort, except as noted. The Government anticipates awarding one or more contracts in accordance with 10 U.S.C 4022, under this Special Topic, but will consider other potential award types listed in the AAL BAA, to include procurement contracts, grants, cooperative agreements, and Other Transaction Agreement. A follow-on production contract or transaction may be awarded to participants that successfully complete a prototype project under this Special Topic.

**PROJECT DESCRIPTION – Background**

The Army is looking for power generation, energy storage, and power distribution capabilities that are modular and interoperable at multiple echelons (organization sizes). Traditional capabilities are not easily distributed from large to small units (Battalion to Squad, for example) and enhancements are needed to meet the increasing power demands of additional battlefield capabilities (such as UAS, cUAS, and EW systems).

Normally, field operations centers serve as power hubs for units from Company to Brigade sizes and run using one or more large (10kW+) and immobile generator(s) which have detectable signatures in acoustic and electromagnetic (EM) domains. Soldier power needs are serviced from these systems, which may create operational challenges due to inflexible output power and limited numbers of connections for power supplies, recharging, etc. In order to provide power for dismounted Soldiers’ equipment, they may carry an excessive weight of device-specific batteries, eliminating the possibility to share power between Soldiers or across devices (such as radios, GPS, sUAS). This problem is exacerbated in Arctic regions where cold temperatures significantly degrade battery performance, resulting in reduced battery operation time and the requirement to carry additional batteries. As a result, commanders must consider power capabilities when planning operations and assessing unit mobility/mission in Arctic environments.

AAL seeks modular plug-and-play systems that seamlessly integrate into a resilient energy network. This includes large generation/storage capabilities down to individual power banks. Solutions should enable interoperable smart hubs, power conversion units, and sharing devices, capable of distributing energy over a 72-hour cycle. Proposed solutions should be capable of interfacing with a variety of standard and non-standard batteries—specifically Conformal Wearable Batteries (CWBs), BB-2590s, and other UAS batteries (XT60 and XT90 connectors)—enabling command centers to efficiently and simultaneously charge or power multiple mission-required technologies while enabling dismounted troops to share power. The Government is also interested in approaches that reduce Soldier weight burden, support low EM/acoustic signature, and enable continuous operations.

**Timeline:** Fieldable Capability Delivery at scale to a BDE for immediate Soldier use is required by November 2026 supporting full-scale testing in Q2FY27.

1) Solution(s) should address at least one level (Soldier, Squad, Platoon, Company to Brigade) and/or combine (scale) to support higher level capability given the following expeditionary performance features/attribute/capabilities:

    a) Architecture: a modular, open system architecture design to allow upgrades of power electronics and energy storage systems as technology improves

    b) Power Output capacity:

        i) Soldier Level – solution must be at least 75 watts (W) (Personal equipment and recharging)

        ii) Squad Level – solution must be at least 300W (Personal equipment and recharging)

        iii) Platoon Level – system solution should be at least 2.0 kilowatts (kW), with a higher peak power capability for short durations (Personal Equipment and recharging)

        iv) Company to Brigade Level - system solution up to 15kW with a higher peak power capability for short durations

    c) Energy storage and distribution systems must not degrade capability by more than 20% at or below -40C.

        i) Soldier Level - storage capacity up to 350 watt-hour (Wh) (scalable modules)

            (1) Input voltage: 5 – 48 VDC and 100-250 VAC 50/60Hz

            (2) Output voltage: 5 – 28 VDC

        ii) Squad Level - storage capacity up to 2.5 kWh (scalable modules)

            (1) Provide at least 1 kW continuous DC or AC power

            (2) Input voltage: 5-48VDC and 100-250 VAC 50/60Hz

(3) Output voltage: 5-48VDC and 100-120 VAC 50/60Hz

iii) Platoon Level - storage capacity up to 4.0 kWh (scalable modules)

(1) Input voltage: 12-48VDC and 100-250VAC 50/60Hz

(2) Output voltage: 12-48VDC and 100-250VAC 50/60Hz

iv) Company to Brigade Level - storage capacity up to 30 kWh (scalable modules)

(1) Input voltage: 12-48VDC and 100-250VAC 50/60Hz

(2) Output voltage: 12-48VDC and 100-250VAC 50/60Hz

d) Transportability: distributive, compact form factor primarily for man-portable transport, sled haul, or mounted on light vehicle(s) when necessary

i) Soldier Level - individual carried, direct physical possession

ii) Squad Level – sled haul

iii) Platoon Level – sled haul or vehicle(s) (varied)

iv) Company to Brigade Level - vehicle(s) (varied)

e) Weight (including all components and accessories)

i) Soldier Level – system weight less than 6 lbs.

ii) Squad Level – system weight less than 15 lbs.

iii) Platoon Level – system weight less than 50 lbs.

iv) Company to Brigade Level – dependent on configuration

f) Interoperability: standard military and/or commercial connectors for inputs and outputs (USB-A/C, hard-wire, 3-prong (grounded) A/C, solar/MC4, XT60) as appropriate for size

g) Communications: monitor and control capabilities using a Controller Area Network (CAN) and/or Tactical Microgrid Standard (TMS) (MIL-STD-3071) (refer to Section 6) for communication as necessary

h) Ruggedization: compliance – meet Ingress Protection, IP67 (water and dust) and MIL-STD-810H (refer to Section 6) or commercial equivalent standard for low temperatures, high relative humidity, shock, vibration, drop, etc.

i) Signature Management: minimization of electromagnetic, thermal, audible, and visual signatures

j) Maintainability: designed for ease of maintenance with readily available components and clear diagnostic capabilities

k) Batteries: interoperate with common military batteries (e.g., BB-2590, etc.) or commercial battery technology in compliance with Section 842 of the FY 2026 National Defense Authorization Act

l) Power Management technology (Hardware and Software):

2) Power management solutions must integrate with unit's existing network and provide insight into unit generators and MIL-STD 3071 (refer to Section 6) compliant power components. The system shall automate measurement, reporting, visualization, and/or forecasting power usage as a sustainment commodity across echelons, while providing an intuitive user interface on existing tactical Command and Control systems.

3) For Applications & Agent solutions, software should be able to aggregate various data sources and data formats to provide real-time situational awareness.

**Scope of the Call**

This special notice invites white papers on the topic of "Expeditionary Power in the Arctic," as part of the Army Applications Laboratory BAA for Disruptive Applications.

A limited number of entities that submit white papers may be selected to submit a full proposal after evaluation of white paper submissions.

The solution may be applicable to various military applications, and it is not known at this time what the quantities or dollar values will be, or which program office or agency may produce any follow-on product.

White papers should be for cutting-edge innovations that could produce discoveries with the potential for significant impacts to enable new and improved Army capabilities regarding power generation, energy storage, power distribution, and power management. Research Themes

Submissions should relate to one or more of the following themes:

1) Small Unit Power Generation and Distribution (Soldier to Platoon) / Portable Batteries Energy Storage

2) Larger / Power Generation (Company to Brigade)

3) Arctic Mobility

4) Power Management technology (Hardware and Software)

**Application Process**

**Eligibility:** Applications for this call will follow the eligibility requirements of the Army Applications Laboratory BAA[1]. Should you require clarity regarding the eligibility of a particular institute, please send a request to one of the contacts detailed in Section 7.

**Number of Awards:** The Government anticipates awarding one or more contracts.

**Duration of Award:** The Government anticipates a 6 to 12-month period of performance.

**Funding:** Funding will be commensurate with the agreed upon scope of work for this effort.

**Evaluators:** Government employees that are subject matter experts in the topic area.

**Closing Date:** The deadline for applications for awards is ***27 MAY 2026 at 11:59 AM CDT***. White papers received after ***11:59 AM CDT on 27 MAY 2026 will not be evaluated or considered*** for award. It is the responsibility of the submitting party to confirm receipt of submissions.

***Note: This special notice will not follow the typical AAL BAA submission instructions that direct white papers to be submitted through the white paper submission tool portal.***

**How to Apply:** For this special notice, white papers must be submitted via email to: arctic-power@aal.mil Please send with the subject line: "Expeditionary Power in the Arctic Special Notice"

Queries relating to the technical, management, contractual, and submission aspects of the call should be addressed to those individuals identified in Section 7. Following evaluation, a select number of companies may be invited to submit a full proposal.

**Security:** Classified Information and Controlled Unclassified Information (CUI) may not be submitted through the submission process identified above. Any classified information or CUI must be coordinated with POCs listed below.

**Evaluation Criteria**

1) Submissions will be evaluated under the evaluation criteria of the AAL BAA. Required documentation is outlined in the BAA document. The following areas will be used during evaluation:

    a) The overall scientific and/or technical merits of the submission and an assessment of technical risk versus beneficial impact of the technology.

    b) The potential contributions of the effort to the Army mission.

    c) Cost realism and reasonableness.

    d) Teaming arrangements and methodologies for creating and modifying teaming arrangements with other organizations (if applicable).

    e) Availability of government funds to complete vendors' proposed R&D activities and likelihood of government funding being available for future procurement and fielding.

    f) The applicant's design development and production capabilities, related experience, facilities, techniques, or unique combinations of these, which are integral factors for achieving the proposed objectives (if applicable).

g) The qualifications, capabilities, and experience of the proposed principal investigator, team leader, or other key personnel who are critical to achievement of the proposed objectives.

h) The applicant's record of past performance.

**Structure of White Paper:**

1) White papers should focus on describing details of the proposed research, including how it is innovative, how it could substantially increase the scientific state of the art, Army relevance, and potential impact.

2) Although the AAL BAA directs white papers to be submitted through the AAL portal, for this special notice, white papers must be submitted via email to: Please send with the subject line: “Expeditionary Power in the Arctic Special Notice”arctic-power@aal.mil Please send with the subject line: “Expeditionary Power in the Arctic Special Notice”

3) Format and content of white papers:

   a) **Cover Page**: The white paper cover page shall include at a minimum:

      i) Title of the white paper

      ii) Name of the organization submitting the white paper

      iii) Names of any participating organizations

      iv) UEI of the organization submitting the white paper

      v) Name, address, email, phone # of the applicant’s POC

   b) **Organization Overview**

      i) Brief description of company (max 500 characters)

      ii) Year Founded

      iii) Company Size (based on full-time employees)

      iv) Investment Status

      v) Approximate Annual Revenue

      vi) (Optional) Other Key Events (funding, patent filing, product launch, awards and recognition, etc.) (Max 80 characters)

      vii) Number of Federal Awards (Non-DoD)

      viii) Approximate Number of Federal Awards (non-DoD)

ix) Number of Federal Awards (DoD)

x) Approximate Number of Federal Awards (DoD)

xi) Company Request (funding, partnership, industry introduction, licensing agreements, other)

xii) SBIR Eligible

c) **Technical Content**:

i) Technical Summary (Max 1,500 characters)

ii) What TRL is your technology in its current form?

iii) What TRL is your technology for your military use case?

TRL info can be found at:

https://www.nasa.gov/directorates/heo/scan/engineering/technology/technology_readiness_level

iv) Tell us more about your technology (technical approach, major milestone and research objectives, ongoing government or commercial work related to proposed solutions, impact to the Army and expected TRL at end of desired effort) (Max 10,000 characters).

v) If selected for a contract, what resources do you require with AAL's assistance.

vi) If selected for a contract, how would a successful AAL engagement impact your company? (max 1000 characters)

d) **Schedule And Cost:**

i) How many months will it take to perform this work?

ii) Approximately how much will it cost to perform this work?

iii) (Optional) If you'd like to propose any cost sharing ideas, please explain (max 250 characters).

e) **Your Team:**

i) Up to three relevant people: Name, Email Phone, City, State, Education Summary, Professional Summary and/or LinkedIn page (Max 500 characters each)

f) **Media Links (Optional)**

i) Technical Specifications

ii) Images

iii) Pitch Decks

g) **Rights in White Papers:**

Government rights **prior** to Contract/Agreement award. By submission of its white paper or proposal, the Submitter agrees that the Government—

May reproduce the paper or proposal, or any portions thereof, to the extent necessary to evaluate the white paper/proposal IAW the evaluation criteria set forth in the Notice, and shall use information contained in the white paper/proposal only for evaluation purposes and shall not disclose, directly or indirectly, such information to any person including potential evaluators, unless that person has been authorized by the head of the agency, his or her designee, or the Contracting/Agreement Officer to receive such information.

**Review**

White papers will be reviewed by the Government evaluators. Companies whose white papers receive the best evaluations will be invited to submit a full proposal.

**Key Dates**

| **Activity** | **Date** |
|---|---|
| Open for white paper submissions | *4 MAY 2026* |
| Webinar<br>(Visit https://aal.mil/expeditionary-power for more information) | *12 MAY 2026* |
| Deadline for white paper submissions | *27 MAY 2026 (11:59 AM CDT)* |
| Anticipated notification of selection to submit a proposal | *12 JUN 2026* |
| Anticipated deadline for full proposal submission | *26 JUN 2026 (11:59 AM CDT)* |
| Anticipated contracts start date | *15 SEPT 2026* |

**References**

1) **Controller Area Network (CAN) Bus** is a common, reliable, and secure physical-layer network protocol used in tactical vehicles and subsystems to connect devices with reduced cabling.

2) **MIL-STD-810H** available at https://quicksearch.dla.mil/qsDocDetails.aspx?ident_number=35978

3) **Tactical Microgrid System (TMS) MIL-STD-3071** available at https://quicksearch.dla.mil/qsDocDetails.aspx?ident_number=285095

4) **Battery Interoperability Standards MIL-STD-3078** available at https://quicksearch.dla.mil/qsDocDetails.aspx?ident_number=285711

5) **Conformable Wearable Battery standards available at https://battery.army.mil/warfighter-hub/conformal-wearable-battery/**

**Contacts**

Below is a list of contact information for additional information related to the call for submissions.

1) **Submission Questions:** arctic-power@aal.mil
    a) **MAJ Jay Greene,** Army Applications Laboratory
2) **Technical Questions:** arctic-power@aal.mil
    a) **Scott Carpenter,** Army Applications Laboratory
3) **Contracting Questions:** arctic-power@aal.mil
    a) **Rudy Estrada** Army Applications Laboratory